SQL Advanced & Expert MCQs – Set 1 (Q1–20)

1. Which join returns only matching rows from both tables?

A) LEFT JOIN
B) RIGHT JOIN
C) INNER JOIN
D) FULL JOIN
Answer: C ✅

---

2. How would you find customers who bought in January but not in February?

A) `INNER JOIN`
B) `LEFT JOIN ... WHERE feb.customer IS NULL`
C) `FULL JOIN`
D) `CROSS JOIN`
Answer: B ✅

---

3. What is a self-join used for?

A) Joining unrelated tables
B) Joining a table to itself
C) Pivoting data
D) Filtering NULL values
Answer: B ✅

---

4. Which query returns the second highest salary?

A) `SELECT MAX(salary) FROM employees`
B) `SELECT salary FROM employees ORDER BY salary DESC LIMIT 2,1`
C) `SELECT salary FROM employees ORDER BY salary DESC OFFSET 1 LIMIT 1`

D) Both B and C
Answer: D ✅

---

5. How can you fetch Nth highest salary without LIMIT/TOP?

A) Using `ROW_NUMBER()`
B) Using `DENSE_RANK()`
C) Using correlated subquery with `COUNT(DISTINCT)`
D) All of the above
Answer: D ✅

---

6. Which performs better for large datasets when checking existence?

A) `IN`
B) `EXISTS`
C) `JOIN`
D) `NOT IN`
Answer: B ✅

---

7. How do you find duplicates in a table?

A) `SELECT * FROM table`
B) `SELECT col, COUNT(*) FROM table GROUP BY col HAVING COUNT(*)>1`
C) `SELECT DISTINCT col FROM table`
D) Using `UNION ALL`
Answer: B ✅

---

8. How do you remove duplicates keeping the latest record?

A) `DELETE WHERE duplicate`
B) Use `ROW_NUMBER() OVER (PARTITION BY col ORDER BY date DESC)`
C) Use `DISTINCT`
D) Use `GROUP BY`
Answer: B ✅

---

9. How do you pivot rows into columns?

A) Using `GROUP BY`
B) Using `CASE WHEN` + `MAX()`
C) Using `JOIN`
D) Using `DISTINCT`
Answer: B ✅

---

10. Which SQL construct is used to unpivot data?

A) `GROUP BY`
B) `CROSS APPLY` with `VALUES()`
C) `DISTINCT`
D) `CUBE`
Answer: B ✅

---

11. Recursive queries are useful for?

A) Aggregations
B) Hierarchical data (e.g., org chart)
C) Pivoting data
D) Indexing
Answer: B ✅

---

12. What is a correlated subquery?

A) A query independent of outer query
B) A query that refers to outer query columns
C) A query used with `DISTINCT`
D) A query using `HAVING`
Answer: B ✅

---

13. How to find employees earning more than their manager?

A) `JOIN employees e ON e.manager_id = m.emp_id WHERE e.salary > m.salary`
B) Using `HAVING salary > manager_salary`
C) Using `GROUP BY manager_id`
D) Using `DISTINCT`
Answer: A ✅

---

14. How to perform FULL OUTER JOIN in MySQL (which doesn't support it)?

A) Use `UNION` of LEFT JOIN and RIGHT JOIN
B) Use `INNER JOIN`
C) Use `CROSS JOIN`
D) Use `SELF JOIN`
Answer: A ✅

---

15. How to get records in one table but not another?

A) `LEFT JOIN ... WHERE right_table.id IS NULL`
B) `INNER JOIN`
C) `FULL JOIN`
D) `CROSS JOIN`
Answer: A ✅

---

16. The consecutive attendance problem (streak) is solved using:

A) `GROUP BY`
B) Window functions with `LAG()`
C) `CROSS JOIN`
D) `EXISTS`
Answer: B ✅

---

17. What is the difference between CROSS APPLY and OUTER APPLY?

A) CROSS APPLY returns only matching rows, OUTER APPLY includes non-matching too
B) Both return same results
C) CROSS APPLY = INNER JOIN, OUTER APPLY = FULL JOIN
D) OUTER APPLY is faster
Answer: A ✅

---

18. How do you calculate a running total?

A) `SUM(col)`
B) `SUM(col) OVER (ORDER BY id)`
C) `GROUP BY col`
D) `DISTINCT SUM(col)`
Answer: B ✅

---

19. The “gaps and islands” problem is typically solved with:

A) Subqueries only
B) Window functions (LAG/LEAD)
C) CROSS JOIN
D) GROUP BY only
Answer: B ✅

---

20. How to calculate percent contribution of each row in total?

A) `(col / SUM(col)) * 100 OVER ()`
B) `(col * 100)`
C) `SUM(col)/COUNT(*)`
D) `AVG(col)`
Answer: A ✅

SQL Advanced & Expert MCQs – Set 2 (Q21–40)

21. Which window function assigns a unique sequential number to rows in a partition?

A) RANK()
B) DENSE_RANK()
C) ROW_NUMBER()
D) NTILE()
Answer: C ✅

---

22. How do you get top N records per group using window functions?

A) Use `ROW_NUMBER() OVER (PARTITION BY col ORDER BY metric DESC)`
B) Use `GROUP BY`
C) Use `DISTINCT`
D) Use `JOIN`
Answer: A ✅

---

23. Which SQL construct calculates a moving average?

A) `AVG(col) OVER (ORDER BY id ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)`
B) `AVG(col)`
C) `GROUP BY col`
D) `LAG()`
Answer: A ✅

---

24. Difference between PARTITION BY and GROUP BY?

A) PARTITION BY aggregates and collapses rows
B) PARTITION BY retains rows while grouping within partitions
C) GROUP BY is faster than PARTITION BY
D) Both are identical
Answer: B ✅

25. Which functions compare values from previous/next rows?

A) ROW_NUMBER()
B) LAG() and LEAD()
C) DENSE_RANK()
D) NTILE()
Answer: B ✅

26. How do you calculate year-over-year growth in SQL?

A) `(current_year - prev_year) / prev_year * 100` using `LAG()`
B) `AVG(col)`
C) `COUNT(*)`
D) `GROUP BY year` only
Answer: A ✅

27. How to get the first and last order per customer?

A) Use `ROW_NUMBER() OVER (PARTITION BY customer ORDER BY order_date ASC/DESC)`
B) Use `DISTINCT`
C) Use `GROUP BY`
D) Use `EXISTS`
Answer: A ✅

28. What does ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW mean?

A) Includes all rows
B) Includes rows from the start of partition up to current row
C) Includes only current row
D) Includes next 2 rows
Answer: B ✅

29. How to calculate running balance in a ledger?

A) `SUM(amount)`
B) `SUM(amount) OVER (PARTITION BY account ORDER BY txn_date)`
C) `AVG(amount)`
D) `ROW_NUMBER()`
Answer: B ✅

---

30. How to find highest and lowest sales per region?

A) Use `MAX()` and `MIN()` window functions partitioned by region
B) Use `DISTINCT`
C) Use `JOIN`
D) Use `GROUP BY` only
Answer: A ✅

---

31. How do you reset ranking for each category?

A) `ROW_NUMBER() OVER (ORDER BY col)`
B) `ROW_NUMBER() OVER (PARTITION BY category ORDER BY col)`
C) Use `DISTINCT`
D) Use `GROUP BY`
Answer: B ✅

---

32. Which function divides rows into equal groups (quartiles, deciles)?

A) RANK()
B) DENSE_RANK()
C) NTILE()
D) ROW_NUMBER()
Answer: C ✅

---

33. Difference between aggregate functions and windowed aggregate functions?

A) Aggregates collapse rows; windowed aggregates keep row-level detail
B) Both collapse rows
C) Both return single values
D) Windowed aggregates are slower
Answer: A ✅

---

34. How do you calculate customer retention (came back next month)?

A) Using `LAG()` on order month and checking consecutive months
B) Using `COUNT(DISTINCT)`
C) Using `JOIN`
D) Using `AVG()`
Answer: A ✅

---

35. Which SQL technique groups user activity into sessions?

A) Sessionization with `LAG()` and time difference checks
B) GROUP BY user_id
C) COUNT(*)
D) CROSS JOIN
Answer: A ✅

---

36. Exclusion joins with window functions are typically implemented by:

A) `ROW_NUMBER()` and filtering
B) CROSS JOIN
C) DISTINCT
D) FULL JOIN
Answer: A ✅

---

37. How can you detect anomalies (e.g., sudden spike in sales) using window functions?

A) Compare row with `LAG()`/`LEAD()` and flag large differences
B) Use `COUNT(*)`
C) Use `JOIN`

D) Use `DISTINCT`
Answer: A ✅

---

38. Running distinct count is computed using:

A) `COUNT(DISTINCT col) OVER (ORDER BY id ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)`
B) Simple COUNT
C) ROW_NUMBER()
D) GROUP BY
Answer: A ✅

---

39. What is the difference between sliding and cumulative windows?

A) Sliding looks at fixed N rows, cumulative grows until current row
B) Both are the same
C) Sliding is faster
D) Cumulative works only with SUM
Answer: A ✅

---

40. How do you calculate weighted average with window functions?

A) `SUM(value * weight) OVER (...) / SUM(weight) OVER (...)`
B) AVG(value)
C) SUM(value)
D) NTILE(4)
Answer: A ✅

SQL Advanced & Expert MCQs – Set 3 (Q41–55)

41. What is the main difference between a CTE and a subquery?

A) Subqueries can be recursive, CTEs cannot
B) CTEs can be recursive and improve readability
C) CTEs are always faster than subqueries
D) Both are identical
Answer: B ✅

---

42. What is one major benefit of using CTEs?

A) They reduce storage size
B) They improve code readability and reusability
C) They automatically create indexes
D) They eliminate joins
Answer: B ✅

---

43. Which query type is best for displaying hierarchical org structures?

A) Subqueries
B) Recursive CTE
C) CROSS JOIN
D) GROUP BY
Answer: B ✅

---

44. How can you calculate factorial using SQL?

A) Recursive CTE multiplying numbers until base case
B) SUM() function
C) GROUP BY
D) LAG()
Answer: A ✅

---

45. How can you generate a calendar table in SQL?

A) Using recursive CTE with dates incremented by 1 day
B) Using GROUP BY
C) Using CROSS JOIN only
D) Using MIN/MAX functions
Answer: A ✅

---

46. How would you implement pagination using CTE?

A) Use `ROW_NUMBER()` in a CTE and filter by row ranges
B) Use GROUP BY
C) Use COUNT()
D) Use DISTINCT
Answer: A ✅

---

47. Can you chain multiple CTEs together?

A) No
B) Yes, by defining them comma-separated before the main SELECT
C) Only in Oracle
D) Only with UNION
Answer: B ✅

---

48. How do you flatten parent-child relationships with CTEs?

A) Recursive traversal until no children remain
B) Using DISTINCT
C) Using GROUP BY
D) Using HAVING
Answer: A ✅

---

49. Difference between recursive CTE and hierarchical query (`CONNECT BY`)?

A) Recursive CTE is ANSI standard, `CONNECT BY` is Oracle-specific
B) Both are identical
C) CONNECT BY is faster everywhere
D) Recursive CTE can't be used for hierarchy
Answer: A ✅

---

50. How can you generate a Fibonacci sequence in SQL?

A) Recursive CTE adding last two numbers until limit
B) Using COUNT()
C) Using GROUP BY
D) Using LAG()
Answer: A ✅

---

51. How do you find gaps in sequential IDs using CTE?

A) Use recursive CTE to generate expected sequence and LEFT JOIN
B) Use DISTINCT
C) Use GROUP BY
D) Use HAVING only
Answer: A ✅

---

52. How do you split a string into rows using SQL?

A) Recursive CTE parsing substrings
B) GROUP BY
C) COUNT()
D) DISTINCT
Answer: A ✅

---

53. How do you detect cycles in a graph using SQL?

A) Recursive CTE with cycle detection (`SEARCH`/`CYCLE` in Oracle, flags in Postgres)
B) GROUP BY
C) COUNT()
D) DISTINCT
Answer: A ✅

---

54. How do you find employees reporting indirectly to a manager?

A) Recursive CTE expanding hierarchy
B) GROUP BY manager_id
C) DISTINCT
D) CROSS JOIN
Answer: A ✅

---

55. What is a common optimization for recursive CTEs?

A) Limit recursion depth with `MAXRECURSION` or base case filters
B) Use COUNT()
C) Replace with CROSS JOIN
D) Always use DISTINCT
Answer: A ✅

SQL Advanced & Expert MCQs – Set 4 (Q56–80)

56. What does `EXPLAIN` show in SQL?

A) Actual runtime performance
B) Query execution plan without running it
C) Data distribution in tables
D) Index statistics only
Answer: B ✅

---

57. What is a clustered index?

A) A separate structure outside the table
B) An index where table data is physically ordered by the key
C) An index for text search only
D) An index that can't be primary key
Answer: B ✅

---

58. What is a covering index?

A) An index that stores all columns of a table
B) An index that includes all columns needed by a query
C) A clustered index
D) A temporary index
Answer: B ✅

---

59. How do you choose the right index?

A) Always index all columns
B) Index frequently filtered/joined columns
C) Index primary keys only
D) Index every numeric column
Answer: B ✅

---

60. Which type of index is fastest for equality lookups?

A) B-Tree
B) Hash
C) Bitmap
D) Composite
Answer: B ✅

---

61. What is index fragmentation?

A) Rows stored sequentially
B) Data pages split, reducing performance
C) Index becoming corrupted
D) Duplicate indexes
Answer: B ✅

---

62. How to avoid full table scans?

A) Use appropriate indexes
B) Use `SELECT *`
C) Use CROSS JOIN
D) Disable indexes
Answer: A ✅

---

63. What is the first step of query execution plan?

A) Sorting results
B) Parsing and generating a logical plan
C) Index scan
D) Joining tables
Answer: B ✅

---

64. How do indexes affect GROUP BY queries?

A) They have no impact
B) They speed up grouping if index matches group columns
C) They slow down grouping
D) They only help DISTINCT queries
Answer: B ✅

**65. Why should you avoid SELECT ?*

A) It reduces readability
B) It loads unnecessary data and hurts performance
C) It can break queries if schema changes
D) All of the above
Answer: D ✅

---

66. How do you detect slow queries in databases?

A) Query logs / slow query logs
B) Execution plans
C) Performance monitoring tools
D) All of the above
Answer: D ✅

67. OLTP vs OLAP query optimization focus?

A) OLTP = fast writes, OLAP = fast reads
B) OLTP = batch queries, OLAP = transactions
C) OLTP = schema-on-read, OLAP = schema-on-write
D) They are the same
Answer: A ✅

68. Best way to optimize JOINs?

A) Always use CROSS JOIN
B) Index join keys
C) Avoid joins
D) Use SELECT *
Answer: B ✅

69. What is a materialized view?

A) A query stored as text only
B) A physical copy of query results stored for faster reads
C) A temporary table
D) A synonym for normal view
Answer: B ✅

70. Difference between partitioning and sharding?

A) Partitioning = within one DB, Sharding = across multiple DBs
B) Both are same
C) Sharding = vertical, Partitioning = horizontal
D) Partitioning = cloud-only
Answer: A ✅

71. What is query hinting?

A) Adding comments to query
B) Forcing the optimizer to choose a specific execution plan
C) Caching results
D) Using DISTINCT
Answer: B ✅

---

72. Why are database statistics important?

A) They improve optimizer decisions for query plans
B) They store indexes
C) They hold table data
D) They are not important
Answer: A ✅

---

73. What is parallel query execution?

A) Multiple queries executed at same time
B) A single query split into parallel tasks across CPUs
C) Running queries asynchronously
D) Using multiple databases
Answer: B ✅

---

74. What is CBO (Cost-Based Optimizer)?

A) Optimizer that uses estimated costs to pick execution plans
B) Optimizer that guesses randomly
C) Optimizer for caching queries
D) Optimizer for distributed DBs only
Answer: A ✅

---

75. When is denormalization beneficial?

A) When minimizing redundancy
B) When improving read performance for analytics
C) When reducing schema complexity
D) Both B and C
Answer: D ✅

---

76. Why does columnar storage improve performance in analytics?

A) It reduces write speed
B) It stores only needed columns, improving scans and compression
C) It duplicates rows
D) It requires less indexing
Answer: B ✅

---

77. What is a filtered index?

A) Index on entire table
B) Index built on subset of rows matching a condition
C) Index built only on primary key
D) Index built across multiple tables
Answer: B ✅

---

78. How do you handle query skew in distributed SQL engines?

A) Broadcast small tables, repartition large tables
B) Always use CROSS JOIN
C) Avoid partitioning
D) Disable parallelism
Answer: A ✅

---

79. What is Adaptive Query Execution (AQE) in Spark SQL?

A) Query plan changes dynamically at runtime based on statistics
B) Static query optimization
C) Manual query hinting
D) Query caching
Answer: A ✅

---

80. Which caching strategy helps SQL performance most?

A) Result caching for repeated queries
B) Always disabling cache
C) Dropping indexes
D) Using SELECT *
Answer: A ✅

🧑‍💻 Part 1: Advanced SQL MCQs (Q1–Q30)

Q1. Which SQL clause is evaluated first in a query?

a) SELECT
b) WHERE
c) FROM
d) GROUP BY

Answer: c) FROM
👉 Execution order starts with `FROM` → `WHERE` → `GROUP BY` → `HAVING` → `SELECT` → `ORDER BY`.

Q2. Which of the following can be used inside a window function?

a) DISTINCT
b) OVER
c) LIMIT
d) TOP

Answer: b) OVER
👉 Window functions require the `OVER()` clause to define partitions and order.

Q3. What does the following SQL return?

```
SELECT COUNT(*)
FROM employees
WHERE salary IS NULL;
```

a) Number of employees with missing salary
b) Number of employees with 0 salary
c) Total employee count
d) Throws error

Answer: a) Number of employees with missing salary
👉 `IS NULL` checks for missing values, not 0.

Q4. Which operator is fastest for checking existence?

a) `IN`
b) `EXISTS`
c) `JOIN`
d) `= ANY`

Answer: b) EXISTS
👉 `EXISTS` stops after the first match, making it faster for large datasets.

Q5. Which query finds the 2nd highest salary?

a) `SELECT MAX(salary) FROM employees;`
b) `SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);`
c) `SELECT TOP 2 salary FROM employees ORDER BY salary DESC;`
d) Both b & c

Answer: d) Both b & c
👉 b works in all DBs, c works in SQL Server (use LIMIT in MySQL/Postgres).

Q6. What does the `WITH RECURSIVE` clause do?

a) Creates temporary table
b) Supports hierarchical queries
c) Replaces joins
d) Optimizes execution

Answer: b) Supports hierarchical queries
👉 Used for tree/graph traversals like employee-manager hierarchy.

Q7. Which one is NOT true about CTEs?

a) They can be recursive
b) They exist only for query execution time
c) They are permanently stored
d) They improve readability

Answer: c) They are permanently stored
👉 CTEs are temporary and vanish after query execution.

Q8. Which aggregate ignores NULLs by default?

a) COUNT(*)
b) SUM(column)
c) AVG(column)
d) Both b & c

Answer: d) Both b & c
👉 `COUNT(*)` counts NULLs, but `SUM` and `AVG` skip them.

Q9. Which index is best for range queries (`BETWEEN`)?

a) Hash Index
b) B-Tree Index
c) Bitmap Index
d) Full-text Index

Answer: b) B-Tree Index
👉 B-Trees are optimized for ordered and range-based lookups.

Q10. Which is the correct syntax for window function ranking?

a) `RANK() GROUP BY()`
b) `RANK() OVER(PARTITION BY col ORDER BY col2)`
c) `RANK() ORDER BY()`
d) `RANK(PARTITION BY col)`

Answer: b) `RANK() OVER(PARTITION BY col ORDER BY col2)`

Q11. Which join may return NULLs from both sides?

a) INNER JOIN
b) LEFT JOIN
c) FULL OUTER JOIN
d) CROSS JOIN

Answer: c) FULL OUTER JOIN

Q12. What does `ROLLUP` do in SQL?

a) Creates pivot tables
b) Adds hierarchical totals in aggregation
c) Drops duplicate rows
d) Joins tables

Answer: b) Adds hierarchical totals in aggregation

Q13. Which query returns employees with salaries greater than their department's average?

a)

```
SELECT name FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees);
```

b)

```
SELECT name FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees WHERE dept_id =
e.dept_id);
```

c) Both

Answer: b) Department-specific average requires correlated subquery.

Q14. Which is not valid in SQL window functions?

a) SUM OVER()
b) ROW_NUMBER OVER()
c) LIMIT OVER()
d) AVG OVER()

Answer: c) LIMIT OVER()

Q15. Which set operator removes duplicates?

a) UNION
b) UNION ALL
c) INTERSECT ALL
d) EXCEPT ALL

Answer: a) UNION

Q16. What does `LATERAL JOIN` do?

a) Joins tables normally
b) Allows subqueries to reference columns from preceding tables
c) Creates recursive query
d) Optimizes join execution

Answer: b) Allows subqueries to reference columns from preceding tables

---

Q17. What's the difference between `RANK()` and `DENSE_RANK()`?

a) Same results
b) RANK skips numbers on ties, DENSE_RANK doesn't
c) DENSE_RANK is faster
d) RANK ignores NULLs

Answer: b) RANK skips numbers on ties, DENSE_RANK doesn't

Q18. Which of these is a valid ACID property?

a) Associativity
b) Consistency
c) Durability
d) Both b & c

Answer: d) Both b & c

Q19. Which SQL keyword returns different values only?

a) DISTINCT
b) UNIQUE
c) GROUP BY
d) ORDER BY

Answer: a) DISTINCT

Q20. What is the default isolation level in most RDBMS (like SQL Server)?

a) SERIALIZABLE
b) READ UNCOMMITTED
c) READ COMMITTED
d) REPEATABLE READ

Answer: c) READ COMMITTED

Q21. Which query finds duplicate rows?

a)

```
SELECT col, COUNT(*)
FROM table
GROUP BY col
HAVING COUNT(*) > 1;
```

b)

```
SELECT DISTINCT col FROM table;
```

c) Both

Answer: a

Q22. What does the following return?

```
SELECT NULL = NULL;
```

a) TRUE
b) FALSE
c) UNKNOWN

Answer: c) UNKNOWN
👉 NULL compared to NULL returns UNKNOWN (use `IS NULL` instead).

Q23. Which index is better for columns with few distinct values?

a) B-Tree Index
b) Bitmap Index
c) Full-text Index
d) Spatial Index

Answer: b) Bitmap Index

Q24. Which of the following statements is true about `CROSS JOIN`?

a) It behaves like INNER JOIN
b) Returns Cartesian product
c) Returns only matching rows
d) Supports ON condition

Answer: b) Returns Cartesian product

Q25. What does `HAVING` do?

a) Filters before grouping
b) Filters after grouping
c) Filters duplicates
d) Optimizes joins

Answer: b) Filters after grouping

Q26. What is the difference between `DELETE` and `TRUNCATE`?

a) DELETE is DDL, TRUNCATE is DML
b) DELETE logs row by row, TRUNCATE is minimally logged
c) DELETE can't have WHERE, TRUNCATE can
d) Both are same

Answer: b

Q27. Which is faster for counting rows in large table?

a) COUNT(*)
b) COUNT(1)
c) COUNT(col)
d) COUNT(DISTINCT col)

Answer: a) COUNT(*)
👉 Optimized internally by most DB engines.

Q28. Which keyword is used for materialized views in SQL?

a) CREATE VIEW
b) CREATE MATERIALIZED VIEW
c) CREATE TEMP VIEW
d) CREATE INDEXED VIEW

Answer: b) CREATE MATERIALIZED VIEW

Q29. Which SQL constraint ensures column values must be unique but allows NULLs?

a) PRIMARY KEY
b) UNIQUE
c) CHECK
d) FOREIGN KEY

Answer: b) UNIQUE

Q30. Which of the following ensures referential integrity?

a) FOREIGN KEY
b) PRIMARY KEY
c) CHECK
d) UNIQUE

Answer: a) FOREIGN KEY

🧑‍💻 Part 3: Advanced SQL MCQs (Q61–Q90)

Q61. Which SQL function returns the first non-NULL value?

a) IFNULL()
b) COALESCE()
c) NVL()
d) All of the above

Answer: d) All of the above (depends on RDBMS)

Q62. Which type of join can produce a Cartesian product?

a) INNER JOIN
b) LEFT JOIN
c) FULL OUTER JOIN
d) CROSS JOIN

Answer: d) CROSS JOIN

Q63. Which of the following is NOT a valid aggregate function?

a) COUNT()
b) SUM()
c) RANK()
d) AVG()

Answer: c) RANK() (it's a window function, not aggregate)

Q64. Which SQL command permanently deletes a table?

a) DROP
b) TRUNCATE
c) DELETE
d) REMOVE

Answer: a) DROP

Q65. What is the purpose of the `NTILE()` function?

a) Divides result set into groups (buckets)
b) Assigns unique row number
c) Removes duplicates
d) Splits strings

Answer: a) Divides result set into groups (buckets)

Q66. Which clause is evaluated last in SQL execution order?

a) ORDER BY
b) SELECT
c) WHERE
d) GROUP BY

Answer: a) ORDER BY

Q67. Which SQL construct enforces that every row in a column must be unique and non-NULL?

a) UNIQUE
b) PRIMARY KEY
c) FOREIGN KEY
d) CHECK

Answer: b) PRIMARY KEY

Q68. Which SQL Server feature captures history of table changes automatically?

a) Triggers
b) Temporal Tables
c) CDC (Change Data Capture)
d) Both b & c

Answer: d) Both b & c

Q69. Which operator is used for regular expressions in PostgreSQL?

a) LIKE
b) ILIKE
c) ~
d) MATCHES

Answer: c) ~

Q70. Which query finds the maximum salary per department?

a)

```
SELECT dept_id, MAX(salary)
FROM employees
GROUP BY dept_id;
```

b)

```
SELECT MAX(salary)
FROM employees;
```

c)

```
SELECT DISTINCT salary
FROM employees;
```

Answer: a

Q71. Which SQL keyword ensures consistent snapshots during a query?

a) LOCK
b) TRANSACTION
c) SNAPSHOT
d) SERIALIZABLE

Answer: d) SERIALIZABLE

Q72. Which clause limits number of rows returned in PostgreSQL/MySQL?

a) LIMIT
b) TOP
c) FETCH FIRST
d) OFFSET

Answer: a) LIMIT

Q73. Which SQL function returns difference between two dates?

a) DATEDIFF()
b) DATEADD()
c) TIMESTAMPDIFF()
d) Both a & c

Answer: d) Both a & c (depends on RDBMS)

Q74. Which of these constraints allows NULL values?

a) PRIMARY KEY
b) FOREIGN KEY
c) UNIQUE
d) Both b & c

Answer: d) Both b & c

Q75. Which clause must be used with GROUP BY to filter aggregated results?

a) WHERE
b) HAVING
c) SELECT
d) DISTINCT

Answer: b) HAVING

Q76. Which SQL feature creates a virtual table from query results?

a) View
b) Index
c) Procedure
d) Trigger

Answer: a) View

Q77. What is the default sort order in ORDER BY?

a) ASC (Ascending)
b) DESC (Descending)
c) Random
d) Depends on RDBMS

Answer: a) ASC (Ascending)

Q78. Which SQL Server function converts string to integer?

a) TO_NUMBER()
b) CONVERT()
c) CAST()
d) Both b & c

Answer: d) Both b & c

Q79. Which SQL statement prevents dirty reads?

a) READ UNCOMMITTED
b) READ COMMITTED
c) SNAPSHOT
d) SERIALIZABLE

Answer: b) READ COMMITTED

Q80. Which SQL keyword combines results and removes duplicates automatically?

a) UNION
b) UNION ALL
c) INTERSECT ALL
d) EXCEPT ALL

Answer: a) UNION

Q81. Which SQL function gets number of rows in a result set?

a) COUNT(*)
b) ROWCOUNT()
c) LEN()
d) SIZE()

Answer: a) COUNT(*)

Q82. Which SQL command is used to rollback a transaction?

a) CANCEL
b) ROLLBACK
c) UNDO
d) REVERT

Answer: b) ROLLBACK

Q83. Which clause allows ordering inside a window function?

a) PARTITION BY
b) ORDER BY
c) GROUP BY
d) HAVING

Answer: b) ORDER BY

Q84. Which SQL operator is used to check if a value matches a list?

a) ANY
b) IN
c) EXISTS
d) BETWEEN

Answer: b) IN

Q85. Which SQL statement ensures transaction changes are saved permanently?

a) SAVE
b) COMMIT
c) END
d) APPLY

Answer: b) COMMIT

Q86. Which of these allows filtering before grouping?

a) WHERE
b) HAVING
c) GROUP BY
d) ORDER BY

Answer: a) WHERE

Q87. Which SQL function extracts part of a string?

a) SUBSTRING()
b) LEFT()
c) RIGHT()
d) All of the above

Answer: d) All of the above

Q88. Which SQL function assigns sequential numbers to rows without ties?

a) RANK()
b) ROW_NUMBER()
c) DENSE_RANK()
d) NTILE()

Answer: b) ROW_NUMBER()

Q89. Which SQL feature enforces domain integrity?

a) CHECK constraint
b) FOREIGN KEY
c) TRIGGER
d) INDEX

Answer: a) CHECK constraint

Q90. Which query finds employees with salary in top 10%?

a) Using `NTILE(10)` over salary
b) Using `PERCENT_RANK()`
c) Using `CUME_DIST()`
d) All of the above

Answer: d) All of the above

Part 4: Advanced SQL MCQs (Q91–Q120)

Q91. Which SQL keyword allows you to rename a column in the result set?

a) CHANGE
b) AS
c) RENAME
d) MODIFY

Answer: b) AS

Q92. Which of the following is true about `EXISTS`?

a) It checks for existence of rows in subquery
b) Returns TRUE if at least one row matches
c) Stops execution after first match
d) All of the above

Answer: d) All of the above

Q93. Which SQL function removes duplicate rows in a query result?

a) DISTINCT
b) UNIQUE
c) GROUP BY
d) TOP

Answer: a) DISTINCT

Q94. Which index type is best for equality lookups?

a) Hash Index
b) B-Tree Index
c) Bitmap Index
d) Full-text Index

Answer: a) Hash Index

Q95. Which of these can cause a deadlock?

a) Multiple sessions locking rows in different order
b) Cross joins
c) Full outer joins
d) HAVING clause

Answer: a) Multiple sessions locking rows in different order

Q96. Which function can find the running average?

a) AVG()
b) AVG() OVER(ORDER BY col ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)
c) RANK()
d) NTILE()

Answer: b) Windowed AVG()

Q97. Which SQL Server feature automatically generates identity values?

a) AUTO_INCREMENT
b) SERIAL
c) IDENTITY
d) NEXTVAL

Answer: c) IDENTITY

Q98. Which of the following is NOT true about PRIMARY KEY?

a) It is unique
b) It can be NULL
c) Only one allowed per table
d) Automatically indexed

Answer: b) It can be NULL

Q99. Which SQL clause returns rows with no matching values in another table?

a) EXISTS
b) NOT EXISTS
c) LEFT JOIN with IS NULL
d) Both b & c

Answer: d) Both b & c

Q100. Which SQL function gives cumulative distribution value between 0 and 1?

a) RANK()
b) DENSE_RANK()
c) PERCENT_RANK()
d) CUME_DIST()

Answer: d) CUME_DIST()

Q101. Which SQL Server command shows query execution plan?

a) EXPLAIN
b) SET SHOWPLAN_ALL ON
c) EXPLAIN ANALYZE
d) TRACE

Answer: b) SET SHOWPLAN_ALL ON (SQL Server), a/c in other DBs

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Q102. Which SQL feature ensures data consistency in concurrent access?

a) Indexes
b) Transactions
c) Views
d) Triggers

Answer: b) Transactions

Q103. Which SQL keyword generates subtotals in GROUP BY queries?

a) GROUPING SETS
b) ROLLUP
c) CUBE
d) All of the above

Answer: d) All of the above

Q104. Which SQL statement resets an identity column?

a) TRUNCATE TABLE
b) ALTER TABLE RESTART IDENTITY
c) DBCC CHECKIDENT
d) Both b & c

Answer: d) Both b & c (depends on DB)

Q105. Which SQL construct runs code automatically after insert/update/delete?

a) Procedure
b) Trigger
c) Function
d) Cursor

Answer: b) Trigger

Q106. Which SQL function extracts year from a date?

a) YEAR()
b) EXTRACT(YEAR FROM date)
c) DATEPART(year, date)
d) All of the above

Answer: d) All of the above (RDBMS-specific)

Q107. Which SQL keyword ensures stable results in non-deterministic queries?

a) ORDER BY
b) GROUP BY
c) DISTINCT
d) LIMIT

Answer: a) ORDER BY

Q108. Which constraint ensures a column must always have a value?

a) DEFAULT
b) UNIQUE
c) NOT NULL
d) CHECK

Answer: c) NOT NULL

Q109. Which query finds duplicate rows with multiple columns?

a)

```
SELECT col1, col2, COUNT(*)
FROM table
GROUP BY col1, col2
HAVING COUNT(*) > 1;
```

b)

```
SELECT DISTINCT col1, col2 FROM table;
```

Answer: a

Q110. Which SQL keyword is used to improve performance by caching execution plans?

a) INDEX
b) HINT
c) PREPARE
d) STORED PROCEDURE

Answer: c) PREPARE

Q111. Which SQL keyword ensures no duplicate rows in UNION?

a) DISTINCT
b) UNIQUE
c) GROUP BY
d) UNION

Answer: d) UNION

Q112. Which SQL Server feature allows partitioning a table?

a) Sharding
b) Partitioned Tables
c) Horizontal Splits
d) Indexed Views

Answer: b) Partitioned Tables

Q113. Which SQL function concatenates strings in PostgreSQL?

a) CONCAT()
b) ||

c) STRING_AGG()
d) All of the above

Answer: d) All of the above

Q114. Which SQL Server isolation level prevents phantom reads?

a) READ COMMITTED
b) REPEATABLE READ
c) SERIALIZABLE
d) SNAPSHOT

Answer: c) SERIALIZABLE

Q115. Which SQL clause limits rows after sorting?

a) OFFSET
b) FETCH NEXT
c) LIMIT
d) All of the above

Answer: d) All of the above (depends on DB)

Q116. Which SQL function generates random numbers?

a) RAND()
b) RANDOM()
c) NEWID()
d) All of the above

Answer: d) All of the above

Q117. Which SQL feature enforces cascading deletes?

a) CASCADE in FOREIGN KEY
b) ON DELETE CASCADE
c) TRIGGER
d) Both a & b

Answer: d) Both a & b

Q118. Which SQL Server command reclaims unused space?

a) VACUUM
b) SHRINKDATABASE
c) ANALYZE
d) OPTIMIZE

Answer: b) SHRINKDATABASE

Q119. Which SQL window function calculates relative rank between 0 and 1?

a) PERCENT_RANK()
b) RANK()
c) DENSE_RANK()
d) NTILE()

Answer: a) PERCENT_RANK()

Q120. Which SQL feature allows defining reusable query blocks?

a) Subquery
b) CTE (WITH clause)
c) Stored Procedure
d) View

Answer: b) CTE (WITH clause)